

FR250SDJ Rear Helper Spring Kit Installation Manual

Model: FR250SDJ (Rear Suspension Package)

Please read these instructions carefully before starting the installation. Ensure all standard safety precautions are observed while working underneath the vehicle.

1. Parts List

Before beginning, verify that all components listed below are present in your kit.

Ref #	Description / Specifications	Qty
1	Suspension Bracket Weldment	2
2	Hollow Rubber Spring	2
3	Hex Cap Screw (HCS) M10 x 40 x 1.5 Gr 8.8 Z	2
4	Shim 1-3/8" L x 1/8" W x 1/32" T with 5/8" Slot	4

2. Installation Instructions

- 1. Remove OEM Bump Stop:** Remove the existing original equipment manufacturer (OEM) bump stop assembly. Retain any OEM fasteners for subsequent re-installation.
- 2. Assemble Helper Spring Unit:** Assemble the hollow rubber spring onto the suspension bracket weldment.
- 3. Install Helper Spring Unit:** Mount the assembled helper spring unit to the vehicle frame. Utilize the existing mounting holes and the retained OEM fasteners or supplied shims as necessary.

Installation Notes:

- Verify that vehicle brake lines or cables do not interfere with the helper spring's position or operation.
- If the vehicle's original equipment includes a peg-style casting, the helper spring is engineered to contact this peg. Do not remove the OEM peg.

3. Clearance Check & Fitting Guidance

- **Clearance Check Reference:** With the vehicle unloaded and at normal ride height, approximately **1/2" to 1-1/2" (13-38 mm)** may be used as a reference range when checking clearance between the rubber spring and its contact point.
- **Larger Gap:** A leveling kit, suspension lift, or other ride-height modification may increase the gap. A larger gap may delay helper spring engagement and reduce support.
- **Smaller Gap:** Too little clearance may cause earlier spring engagement and may result in a firmer unloaded ride.
- **Under Load:** Cargo, accessories, or vehicle load may reduce or eliminate the gap as the helper spring engages.
- **Installation Check:** Actual clearance may vary with vehicle configuration, ride height, suspension condition, and load. If the unloaded gap is significantly outside this reference range, verify ride height, suspension modifications, load condition, and correct installation.

Important Safety Warnings:

- Do not exceed the vehicle manufacturer's recommended axle weight rating / GAWR.
- Inspect for potential interference between the hollow rubber spring and adjacent vehicle components. Under load, the rubber spring will expand in diameter.

Note: Illustrations are for reference only. Vehicle configuration and actual component appearance may vary. Install according to the vehicle-specific mounting points and standard mechanical practices.

INSTALLATION DIAGRAM

